

1. S. C. Gupta and V. K. Kapoor. Fundamentals of Applied Statistics. Sultan Chand and Sons.
2. Benjamin B, Health and Vital Statistics, Allen and Unwin

Programme	B. Sc. Statistics				
Course Code	STA6EJ305				
Course Title	Actuarial Statistics				
Type of Course	Major Elective				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					

Course Summary Objective	To learn the life tables used in insurance products. To learn the concept of interest, different life insurance products, life annuities, net premiums. To motivate students to prepare for exams required for employment in the actuarial science profession.
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Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Discuss future life time distributions and their probabilities.	U	C	Instructor-created exams / Quiz
CO2	Know the concept of life table.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Apply different kinds of interest rates expressed in different time periods.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Understand the basics life assurance and life annuity contracts.	U	C	Instructor-created exams / Home Assignments
CO5	Understand the utility theory, insurance products and life tables. Understand the concept of interest.	Ap	P	One Minute Reflection Writing assignments
CO6	: Understand the concept of life insurance and the existing insurance products of different insurance company. Know life annuities, net premium and net premium reserves	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Future life time distribution		10	15
	1	Future life time random variables,		
	2	Force of mortality, Laws of mortality		
	3	De Moivre's law, Gompertz's Law (Definition only)		
	4	Makeham's Law, Weibull's Law (Definition only)		
	5	Probabilities of survival and death, Curtate Future life time		
	Sections from References:			
II	Life Tables		12	20
	6	Construction of a life table		
	7	Assumptions for fractional ages		
	8	Uniform distribution of deaths		
	9	Balducci assumption,		
	10	Constant force of mortality assumption		
	11	Select and ultimate life tables		
	Sections from References:			
III	Rates of interests and Annuities		16	20
	12	Compound interest and discount factor		
	13	Nominal rate of interest		
	14	Force of interest		
	15	Accumulated value		
	16	Annuities		
	17	Annuities certain- Immediate and due		
	18	monthly annuity certain		
	19	Continuous annuity certain		
	20	Deferred annuity		
	Sections from References:			